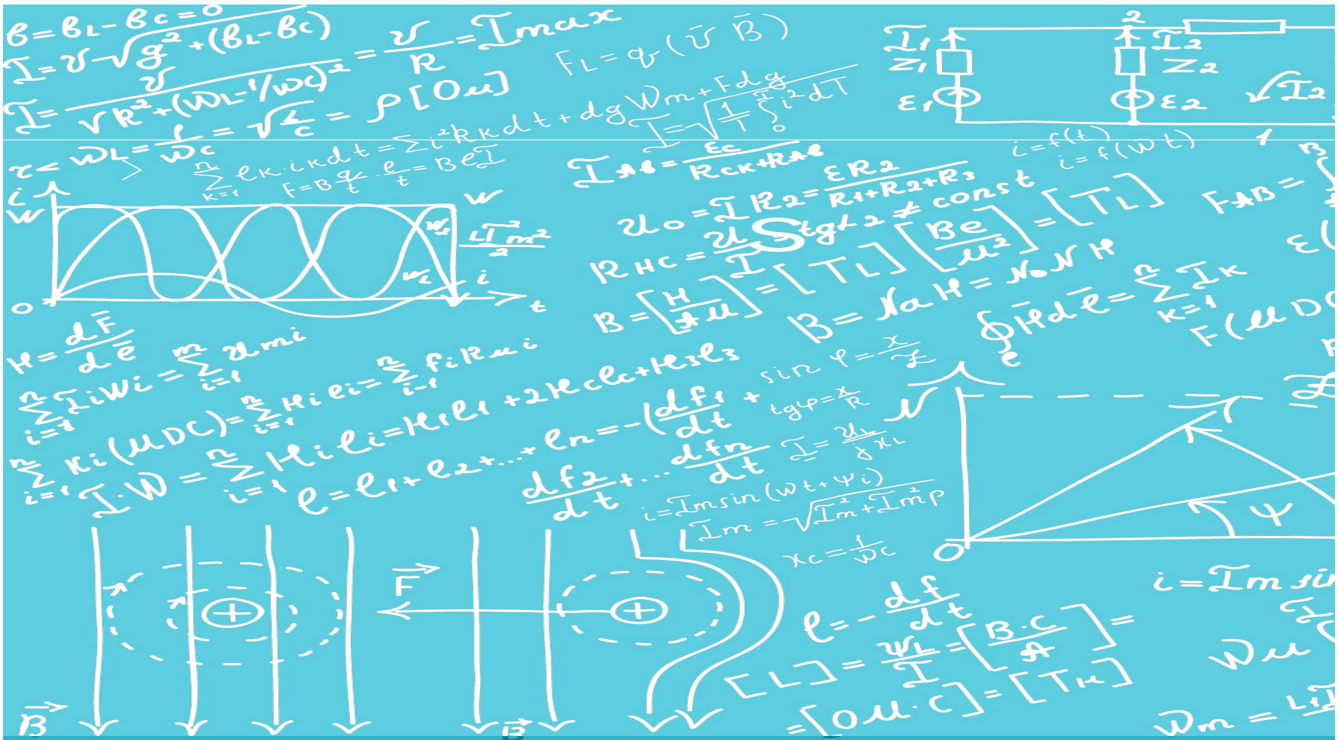

Chapter P2

Practical skills at A Level

LEARNING INTENTIONS

In this chapter you will learn how to:

- develop a systematic approach to carrying out experiments, including planning, setting up apparatus, investigating and recording results, analysing data and writing conclusions
- plan an investigation to test a relationship or investigate a problem, identifying the dependent, independent and control variables
- use logarithms and logarithmic graphs
- combine uncertainties, extending work from Practical Skills at AS Level
- plot error bars on graphs and find uncertainties in gradients and intercepts.

BEFORE YOU START

Do you know how to:

- estimate an uncertainty?
- tell the difference between systematic, mean and random errors?
- present data in a suitable table?
- draw a simple graph with suitable axes and labels?
- describe a simple experiment giving the steps logically one after the other?
- identify simple problems with an experiment and suggest changes to improve an experiment?
- use logarithms to base e and base 10?

P2.1 Planning and analysis

The practical work in the second year of your A level course builds on what you have covered in the first year. Tests and examinations you may take during your studies will ask you to demonstrate your abilities in two key areas:

- planning experiments
- analysis and evaluation of your results, including any conclusions you can draw.

Why do you think that experimental work is so important and has made such a difference to modern physics? What can you do to improve your experimental technique?

In this chapter, we will look at the different skills that you need to demonstrate your practical abilities.

P2.2 Planning

As you progress through your A level physics studies, you should think about and continually develop your approach to planning experiments. The experiments you will be asked to plan by your teacher will usually provide you with a scenario and sometimes a relationship or an equation that you are to use and test. Often, particular items of apparatus are mentioned and you should use these items, even if you think there is a better method. Sometimes, the experiment will seem familiar to you and sometimes it will be completely new. Before you start, it is important to read the scenario carefully. It is also important to read, understand and re-read any parts of a question you need to answer, before starting on your plan.

In producing your plan, you should draw a diagram showing the actual apparatus to be used, and pay particular attention to the:

- procedure to be followed
- measurements to be taken
- control of variables
- analysis of the data
- safety precautions to be taken.

Defining the problem: identifying the variables

It may seem obvious, but the first thing is to identify the problem. To do that you must identify the:

- **independent variable** in the experiment
- **dependent variable** in the experiment
- **control variables** (the quantities that are to be controlled or kept constant).

It is usually a good idea to start with a clear statement about the variables as the first part of your plan.

Here is an example of the sort of problem you might face in planning an experiment.

The deflection of a balloon by a jet of air is shown in Figure P2.1. You need to plan an investigation to show that tan θ is inversely proportional to v^2, where θ is the angle between the ground and the string of the balloon and v is the speed of the air hitting the balloon. You are unlikely to have seen this experiment before, but this should not concern you.

In this example, the speed v of the air is the variable that you will need to alter and so this is the independent variable; the angle θ is the variable that changes as a result and so this is the dependent variable.

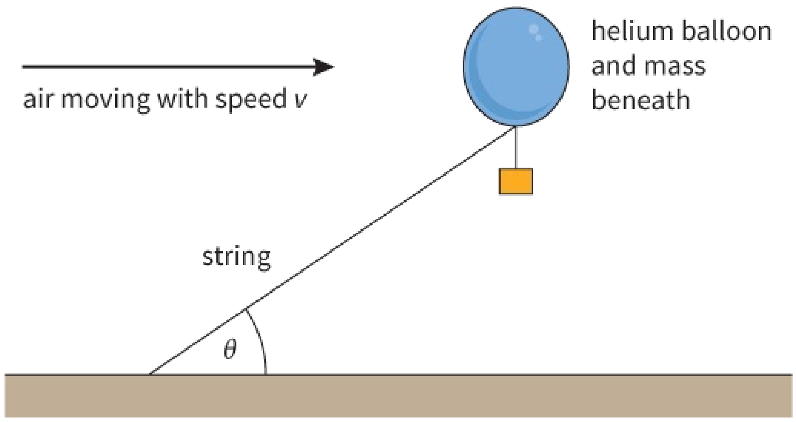

Figure P2.1: A balloon is deflected as the air moves at different speeds.

But what quantities are kept constant? These are the quantities that are controlled. You may be able to think of many, such as the total mass of the balloon and the mass placed underneath it. This total mass is certainly one quantity that should be kept constant, but it is not something that is likely to change during the course of the experiment. In terms of planning the experiment, you need to think about quantities that may easily change during the experiment if care is not taken. For example, you might realise that:

- the balloon may be deflected downwards, particularly if the air blows more strongly; then the air will hit the top, rather than the middle, of the balloon
- the balloon may warm up and expand in size; then more air will hit the balloon.

In either case, the experiment will then not just be testing the effect of the air speed. Your plan should clearly state what you need to keep constant. In our example:

- make sure that the jet of air is always horizontal and hits the middle of the balloon
- keep the temperature of the air inside the balloon constant.

As you can see, you need to think carefully about the experiment. Avoid giving wrong suggestions, for example, keeping the length of the string constant. If the string is longer the balloon may be out of the jet of air, and so it is not entirely a wrong suggestion, but it is not a primary quantity to be kept constant.

Question

1 An experiment is being planned to measure how the resistance of a wire depends on the cross-sectional area of the wire. What are the independent and dependent variables? Suggest three quantities that might be controlled.

Methods of data collection

The next task is to think about how you are going to carry the experiment out. Once you have a method in mind, you should be able to describe:

- the method to be used to vary the independent variable
- how the independent variable is to be measured
- how the dependent variable is to be measured
- how other variables are to be controlled
- the arrangement of apparatus for the experiment and the procedures to be followed, with the aid of a clear, labelled diagram.

It may be worthwhile jotting down your thoughts about the experiment on a rough piece of paper before you start, but do make sure that you write up all your points. It is particularly important to say what you actually measure and how the measurement is made. It may seem obvious to you that a particular quantity is measured, but unless you write it down it is not part of your plan.

Always check that in your account you have clearly said what you will:

- measure and how you will measure it
- change and how you will change it
- keep constant and how this is achieved.

Let's use the example of the balloon deflected by a current of air to show how you could approach this part of the plan.

Describing the experiment

First, describe how to change the independent variable and state what instrument is used to measure it. The apparatus shown in Figure P2.1 does not help very much and you must use your general knowledge and suggest, for example, that a wind fan be used. To change v will mean either changing the distance from the fan to the balloon or adjusting the power supply voltage to the fan.

You will also a need a wind speed indicator, sometimes called an anemometer, to measure the independent variable. Perhaps you have never seen or used a wind speed indicator, but clearly there must be an instrument to actually measure v. You may have to think very carefully to find a sensible instrument when the quantity is unusual.

The instrument to measure the dependent variable is much simpler – a protractor – although it may have to be a large protractor. Alternatively, you could use a ruler to measure the height h from the bench to the top of the string and the length l of the string, and then use $\sin\theta = \frac{h}{l}$ to find θ.

At this stage, try to suggest how to keep at least one of your 'controlled quantities' constant. For example, for the suggestions made earlier:

- compensate for deflection of the balloon downwards by a faster wind by lowering the fan, so that air from the fan stays horizontal and is always aimed at the centre of the balloon.
- keep the temperature of the air inside the balloon constant by leaving the balloon in a room with constant temperature for many hours before the experiment starts, and ensuring that the fan used blows air from the room.

As you can see, you have to think carefully about what happens during the experiment.

As you now have a clear idea of the experiment in your mind, draw a labelled diagram showing everything that you have mentioned. In this example, you could draw the fan, possibly its supply, a protractor, and even an anemometer.

Now describe your planned experiment, making sure that you describe a logical sequence of steps to follow. If you find this difficult, a labelled diagram of each step can sometimes be useful. For example, you might draw a diagram where you remove the balloon and put your wind speed measurement device in place of the balloon to measure v. Did you realise that the reading for v must be made exactly where the balloon was placed?

Additional details

It is also helpful to give additional details. In particular, make sure you suggest anything that needs to be done to ensure there is a large change in the dependent variable.

In the experiment with the balloon, you need a large change in θ as v changes. The readings would not be useful if θ was always very close to one value, for example, 90°. How can this be achieved?

Obviously, the largest air speed must be strong enough to cause a significant deflection. If the deflection is too small, then the mass under the balloon can be decreased; if it is too large, then the mass can be increased. It might be sensible to have the air speed as large as possible and adjust the mass under the balloon until θ is about 30°, and then check that θ varies from 30° to 90° as the fan is slowly moved further away. Of course, the mass under the balloon is then kept constant.

You might also think of any difficulties in carrying out the experiment. For example, draughts must be avoided and you must wait until the balloon has stopped swinging before taking a reading of θ.

Safety

It may seem strange, but you should always comment on safety when asked to carry out any experiment. In some situations, the risks may be unimportant, and it may be sufficient to mention simple ideas such as wearing goggles to protect the eyes when heating liquids or when handling stretched wires, using a safety screen, ensuring that the apparatus is stable and not easily knocked over, using a sand tray under heavy weights to make sure that weights do not fall on your foot and switching off currents when not in use so that wires do not overheat.

In our example with the balloon, keeping away from the rotating blades in the fan and wearing goggles to avoid air blowing into your eye should be sufficient. Do give some detail in your suggestions and do not just say 'use goggles'.

WORKED EXAMPLE

1 Plan an experiment to measure the resistivity ρ of glass, which is about 10^{10} Ω m. You have available a number of sheets of glass of the same size but with different thicknesses. Resistivity ρ is defined as $\rho = \frac{RA}{l}$.

Step 1 Identify the variables.

- The independent variable is the thickness l of the glass.
- The dependent variable is the resistance R of the glass. Finding R involves measuring the p.d. across the glass and the current in the glass.
- The control variable is the area A of the glass. Since this is mentioned in the question, suggest also that the temperature must be constant.

Step 2 Describe the method of data collection in logical steps.

To alter the independent variable, use glass sheets of different thickness but the same area. The thickness of each piece of glass is measured with a micrometer at several places and averaged.

The area A is required. This can be found by measuring the length and breadth of each sheet with a rule and multiplying the values together.

Draw a circuit diagram of an ammeter in series with the glass sheet and a power supply, with a voltmeter across the glass. Connections are made to the large surfaces of the glass. This can be done using aluminium foil, or metal plates as in a capacitor, which closely touch each large face of the glass sheet. Use a diagram to show how this is done.

The logical steps are then to record ammeter and voltmeter readings with one thickness of glass. Then repeat the readings with different thicknesses, suggesting sensible thicknesses of glass, perhaps every mm from 1 mm to 10 mm. If you are going to perform the experiment these thicknesses may be available, but if you are merely planning the experiment then you must suggest sensible values.

Step 3 Add any additional details. How can you obtain reasonable values? Think about the size and thickness of the glass to be used and whether you can detect a reasonable change in the dependent variable, the resistance. You might, for example, suggest using a sheet of glass 1 m^2 in area and 1 mm thick. Its resistance is then:

$$\begin{aligned} R &= \frac{\rho l}{A} \\ &= 10^{10} \times 0.001 \\ &= 10^{7}\,\Omega \end{aligned}$$

Can this be measured with ordinary laboratory apparatus? What voltages and what meters are suitable? A voltage of 10 V produces a current of 1 μA, which is measurable, but 100 V gives a current of 10 μA, which may be easier to measure but more dangerous. With glass of thickness between 1 and 10 mm the current will be 1 to 10 μA and so the ammeter should measure from 1 to 10 μA or up to 10 μA.

As you can see, this means that you need some idea of the size of quantities that can be measured. In this example, you need to know what currents and voltages can be measured with ordinary laboratory equipment.

You may also give additional detail by describing how to attach the metal foil as contacts onto the large faces of the glass sheet with weights on top, or suggest that the glass be cleaned and dried.

Step 4 State any safety points. Glass can cut a person's skin and so gloves should be worn. If voltages above about 50 V are to be used, then use rubber gloves to avoid an electric shock or cover all exposed metal parts with insulation.

Step 5 Give your method of analysis. Remember, every derived quantity must be explained, so do not forget to state that for each thickness the voltage and current readings are used to find the resistance with the formula $R = \frac{V}{T}$.

Since $R = \frac{\rho l}{A}$, choose to plot a graph with R on the y-axis and l on the x-axis. The graph should be a straight line through the origin – a diagram may help here.

The gradient of the graph is $\frac{\rho}{A}$, so ρ = gradient × A.

Questions

2 What other graph can be plotted in Worked example 1, and how is its gradient used to find ρ?

3 A nail is placed with its sharp end just touching a piece of wood. When a mass falls with a velocity v and hits the nail, it drives the nail into the wood. It is suggested that the depth d that the nail moves into the wood is related to v by the equation $d = kv^2$, where k is a constant.

a Suggest:

i the independent, dependent and control variables

ii how the velocity of the falling mass can be measured as it hits the nail

iii sensible values for d and how they may be achieved and measured

iv the graph to be plotted and what it shows if the relationship is true.

b Write a logical step-by-step method to test the relationship.

P2.3 Analysis of the data

Whether you are dealing with data you have collected in an experiment, or data provided to you, you will need to analyse it. You need to describe how the data is used in order to reach a conclusion, and give details of any derived quantities that are calculated.

First, look carefully at the quantities in the relationship you have suggested (or at the formula that may be suggested when you are given an experiment to carry out). In our example with the balloon, tanθ is inversely proportional to v^2, which means that the formula is $\tan\theta = \frac{k}{v^2}$, where k is a constant.

If possible, you should suggest plotting a graph that you know is a straight line if the equation is correct. In our example, since the equation for a straight line is $y = mx + c$, the y-axis of the graph should be tanθ and the x-axis should be $\frac{1}{v^2}$.

You must clearly state:

- what is plotted on each axis of your graph
- that the relationship is valid if the graph gives a straight line through the origin.

You may prefer to draw a sketch graph to show what you mean, but always state clearly what type of graph you are going to use.

More complicated analysis of data

In P1 Practical skills at AS Level, we saw how to interpret equations of the form $y = mx + c$ and how to use a straight-line graph to find the constants m and c. However, you also need to be able to deal with quantities related by equations of the form $y = ax^n$ and $y = ae^{kx}$. For these, you need to be able to use logarithms (logs).

There are two common types of logarithm (see Chapter 20). The first type is sometimes called a natural logarithm, or a logarithm to base e, and is written as ln. The second type is a logarithm to base 10 and is written as lg. The ln type is more useful when dealing with an exponential formula such as e^{kx} but, otherwise, either type may be used. Look closely at any question to see which type is used. Do not mix the different types together in the answer to one question.

The unit of a quantity involving logarithms is specified in an unusual way. For example, the natural logarithm of a quantity s measured in metres is written as ln (s / m) and not as ln (s) / m or ln (s) / ln (m). You can see that the unit is written inside the bracket with the quantity.

You need to be able to take logarithms of equations of the form $y = ax^n$ and $y = ae^{kx}$. (Recall that an equation remains balanced if the same operation is performed on each side.)

Consider the equation: $y = ax^n$

Taking logarithms of both sides gives:

$$\lg y = \lg a + n \lg x$$

$$\ln y = \ln a + n \ln x$$

Now consider the equation: $y = ae^{kx}$

Taking logarithms of both sides gives:

$$\ln y = \ln a + kx$$

(To obtain these results, we have used the rules for logarithms that you ought to know:

log of a product	$\log(ab) = \log(a) + \log(b)$
log of a ratio	$\log\left(\frac{a}{b}\right) = \log(a) - \log(b)$
log of a power	$\log(a^n) = n\log(a)$

Questions

4 Calculate:

a lg 10

b ln 10

c lg 100

d lg 5

e the antilogarithm to base 10 of 1 (i.e., find x where $\lg x = 1$)

f the antilogarithm to base e of 0.5 (i.e., find x where $\ln x = 0.5$).

5 The number $48 = 3 \times 2^4$. Calculate lg 48 and lg 3 + 4 lg 2. Why are they the same?

Which graph to plot?

In handling data, our aim is usually to process the data to obtain a straight line graph. Then we can deduce quantities from the gradient and the intercepts. Table P2.1 shows graphs that can be plotted for different relationships, and the quantities that can be deduced from the graphs.

Relationship	**Graph**	**Gradient**	**Intercept on *y*-axis**	**because …**
$y = mx + c$	y against x	m	c	
$y = ax^n$	$\ln y$ against $\ln x$ $\lg y$ against $\lg x$	n	$\ln a$ $\lg a$	$\ln y = n \ln x + \ln a$ $\lg y = n \lg x + \lg a$
$y = ae^{kx}$	$\ln y$ against x	k	$\ln a$	$\ln y = kx + \ln a$

Table P2.1: Choice of axes for straight-line graphs.

A relationship of the form $y = ax^n$

A ball falls under gravity in the absence of air resistance. It falls a distance s in time t. The results are given in the first two columns of Table P2.2. A graph of distance fallen against time gives the curve shown in Figure P2.2.

Time t / s	**Distance fallen s / m**	**ln (t / s)**	**ln (s / m)**
0.20	0.20	−1.61	−1.61
0.40	0.78	−0.92	−0.25
0.60	1.76	−0.51	0.57
0.80	3.14	−0.22	1.14
1.00	4.90	0.00	1.59
1.20	7.05	0.18	1.95

Table P2.2: Results for a ball falling under gravity.

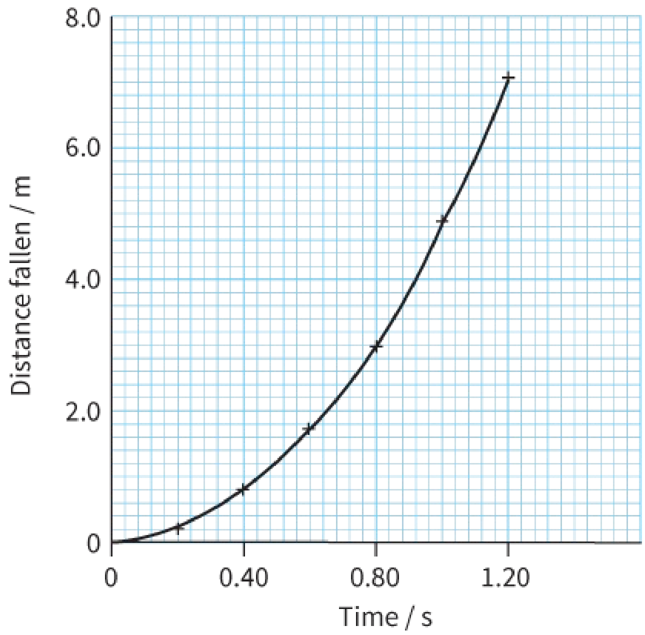

Figure P2.2: A distance–time graph plotted using the data in Table P2.2.

Because this is a curve, it tells us little about the relationship between the variables. If, however, we suspect that the relationship is of the form $y = ax^n$, we can test this idea by plotting a graph of ln *s* against ln *t* (a 'log–log plot'). Table P2.2 shows the values for ln *s* and ln *t*, and the resulting graph is shown in Figure P2.3. (Notice that here we are using natural logs, but we could equally well use logs to base 10.)

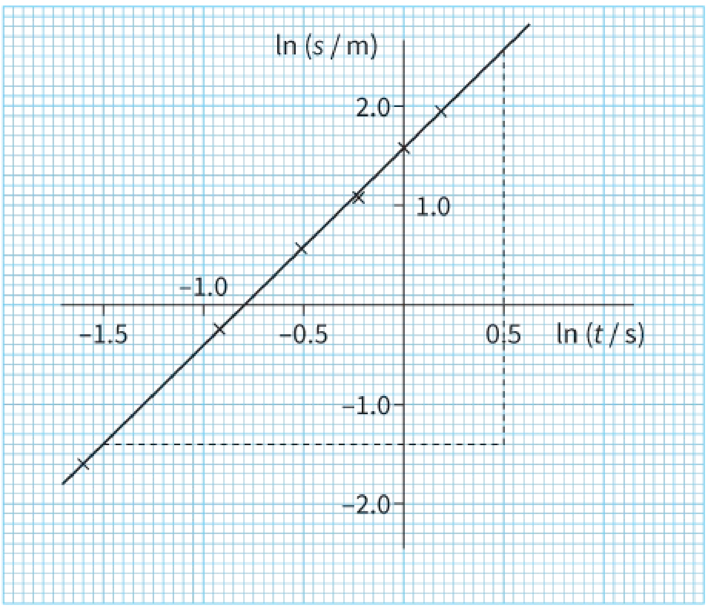

Figure P2.3: A log–log plot for the data shown in Table P2.2.

Because the graph is a straight line, the relationship must be of the form $y = ax^n$ But what are the values for *a* and *n*?

From the graph, the gradient is equal to the value of *n*, the power of *t*:

$$\begin{aligned} n &= \text{gradient} \\ &= \frac{(2.55-(-1.4))}{(0.5-(-1.5))} \\ &= \frac{3.95}{2.0} \\ &= 1.98 \approx 2.0 \end{aligned}$$

So the equation is of the form $s = at^2$. The intercept on the y-axis is equal to ln a, so:

$$\ln a = 1.6$$

By taking the antilogarithm we get:

$$a = 4.95 \text{ m s}^{-2} \approx 5.0 \text{ m s}^{-2}$$

If we think of the equation for free fall $= \frac{1}{2}gt^2$, the constant $a = \frac{1}{2}g$. But $g = 9.8$ m s^{-2}, which is consistent with the value we get for our constant.

A relationship of the form $y = ae^{kx}$

A current flows from a charged capacitor when it is connected in a circuit with a resistor. The current decreases exponentially with time (the same pattern we see in radioactive decay).

Figure P2.4 shows the circuit and Table P2.3 shows typical values of current I and time t from such an experiment.

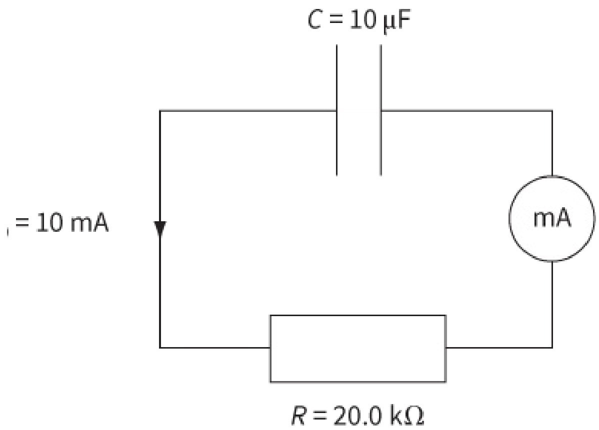

Figure P2.4: A circuit for investigating the discharge of a capacitor.

Current *I* / mA	**Time *t* / s**	**ln (*I* / mA)**
10.00	0.00	2.303
6.70	0.20	1.902
4.49	0.40	1.502
3.01	0.60	1.102
2.02	0.80	0.703
1.35	1.00	0.300

Table P2.3: Results from a capacitor discharge experiment.

The graph obtained from these results (Figure P2.5) shows a typical decay curve, but we cannot be sure that it is exponential. To show that the curve is of the form $I = I_0e^{kt}$, we plot ln I against t (a 'log-linear plot'). Values of ln I are included in Table P2.3. (Here, we must use logs to base e rather than to base 10.)

The graph of ln I against t is a straight line (Figure P2.6), confirming that the decrease in current follows an exponential pattern. The negative gradient shows exponential decay, rather than growth.

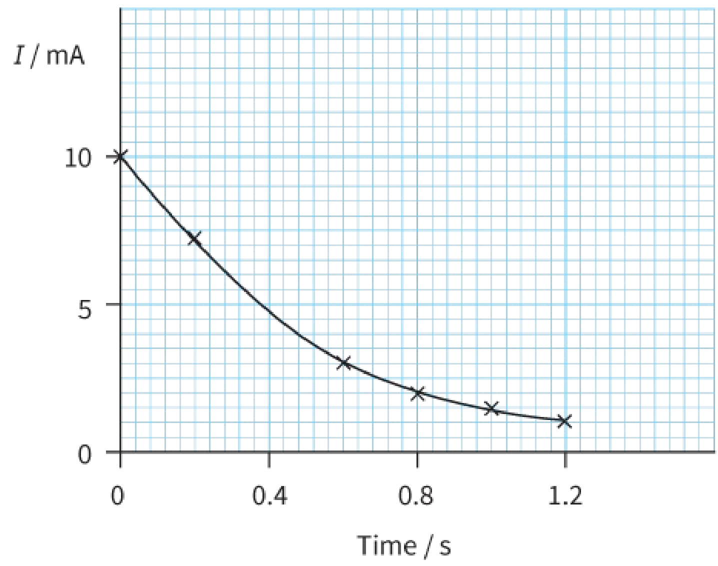

Figure P2.5:

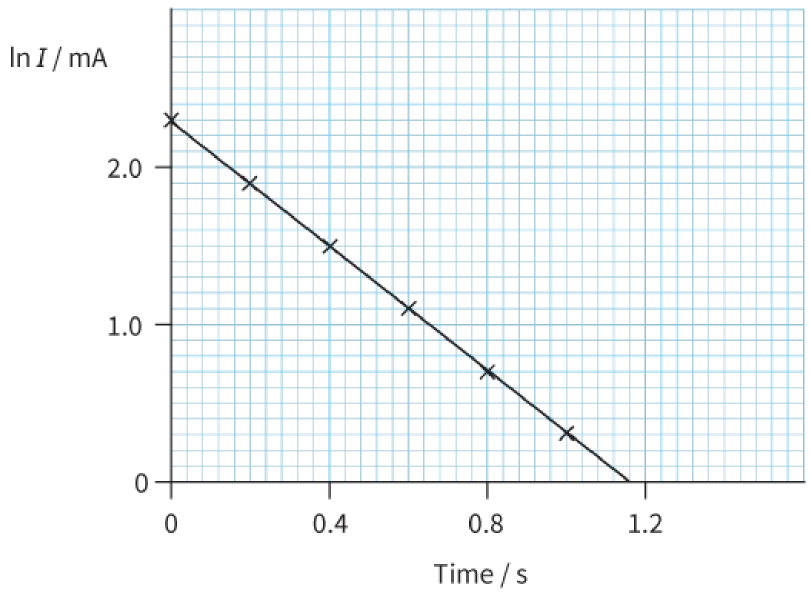

Figure P2.6:

The gradient of the graph gives us the value of the constant k:

$$\begin{aligned} k &= \text{gradient} \\ &= \frac{(0-2.30)}{(1.16-0)} \\ &= -1.98\text{s}^{-1} \approx -2.0\ \text{s}^{-1} \end{aligned}$$

From the graph, we can also see that the intercept on the y-axis has the value 2.30 and hence (taking the antilogarithm) we have $I_0 = 9.97 \approx 10$ mA. Hence, we can write an equation to represent the decreasing current as follows:

$$I = 10\ \mathrm{e}^{-2.0t}$$

We could use this equation to calculate the current at any time t.

Questions

6 In the expressions that follow, x and y are variables in an experiment. All the other quantities in the expressions are constants.

In each case, state the graph you would plot to produce a straight line. Give the gradient of each line in terms of the constants in the expression.

a $y = kx^{\frac{3}{2}}$

b $y = cx^q$

c $m = \frac{8x}{By^2}$

d $y = y_0 e^{kx}$

e $R = \frac{(y - y_0)}{x^2}$

7 The period of oscillation T of a small spherical mass supported by a length l of thread is given by the expression:

$$T = 2\pi\sqrt{\frac{l}{g}}$$

where g is the acceleration due to gravity.

Design a laboratory experiment using this expression to determine the acceleration due to gravity. You should draw a diagram showing the arrangement of your equipment. In your account, you should pay particular attention to:

- the procedure to be followed
- the measurements to be taken
- analysis of the data to determine g
- any safety precautions that you would take.

P2.4 Treatment of uncertainties

All results should include an estimate of the absolute uncertainty. For example, when measuring the time for a runner to complete the 100 m you may express this as (12.1 ± 0.2) s. This can also be expressed as a percentage **uncertainty** (see Chapter P1); the percentage uncertainty is equal to $\frac{0.2}{12.1} \times 100\% = 1.655$, so we write the value as 12.1 s ± 1.7%, or even 12.1 s ± 2%.

Combining uncertainties

You should read through again the topic in Chapter P1 on combining uncertainties and also how to measure uncertainty if you are not sure.

Uncertainties and graphs

We can use error bars to show uncertainties on graphs. Table P2.4 shows results for an experiment on stretching a spring.

Load / N	**Length of spring / cm**	**Extension / cm**
0	12.4 ± 0.2	0.0
1.00	14.0 ± 0.2	1.6 ± 0.4
2.00	15.8 ± 0.2	3.4 ± 0.4
3.00	17.6 ± 0.2	5.2 ± 0.4
4.00	18.8 ± 0.2	6.4 ± 0.4
5.00	20.4 ± 0.2	8.0 ± 0.4

Table P2.4: Results from an experiment on stretching a spring.

When plotting the graph, the points are plotted as usual, and then they are extended to show the maximum and minimum likely values, as shown in Figure P2.7. Then the best fit line is drawn.

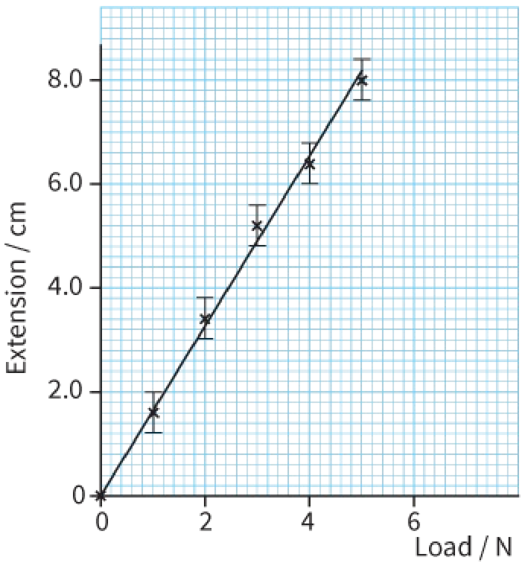

Figure P2.7: A graph representing the data in Table P2.4, with error bars and a line of best fit drawn.

To estimate the uncertainty in the gradient, we draw not only the best fit line but also a **worst acceptable line**, passing through the extremes in the error bars as shown in Figure P2.8.

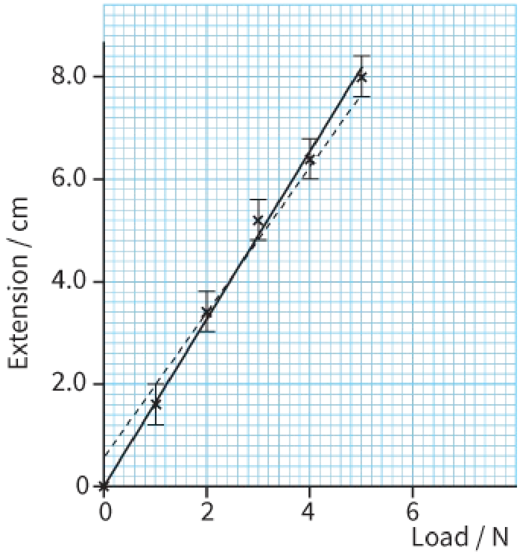

Figure P2.8: The same graph as in Figure P2.7, with a 'worst acceptable' line drawn (dashed).

The gradients for both best fit and worst fit lines are calculated and the uncertainty is the difference in their gradients:

uncertainty = (gradient of best fit line) − (gradient of worst acceptable line)

In our experiment, the gradients are:

line of best fit:

$$\begin{aligned} \text{gradient} &= \left(\tfrac{8.2-0}{5.0-0}\right) \text{cm N}^{-1} \\ &= 1.64 \text{ cm N}^{-1} \approx 1.6 \text{cm N}^{-1} \end{aligned}$$

line of worst fit:

$$\begin{aligned} \text{gradient} &= \left(\tfrac{7.6-0.6}{5.0-0}\right) \text{cm N}^{-1} \\ &= 1.4 \text{ cm N}^{-1} \end{aligned}$$

So the uncertainty in the gradient = 1.6 − 1.4 = ± 0.2 cm N^{-1}

The gradient is therefore: 1.6 ± 0.2 cm N^{-1}.

Uncertainties and logarithms

When a log graph is used and we need to include error bars, we must find the logarithm of the measured value and the logarithm of either the largest or the smallest likely value. The uncertainty in the logarithm will be the difference between the two.

WORKED EXAMPLE

2 The resistance of a resistor is given as (47 ± 5) Ω. The value of ln (R / Ω) is to be plotted on a graph. Calculate the value and uncertainty in ln (R / Ω).

Step 1 Calculate the logarithm of the given value:

ln (R / Ω) = ln 47 = 3.85

Step 2 Calculate the logarithm of the maximum likely value:

maximum likely value = 47 + 5 = 52 Ω

ln 52 = 3.95

Step 3 The uncertainty is the difference between the two logarithms:

uncertainty in ln R = 3.95 − 3.85 = 0.10

Thus, ln (R / Ω) = 3.85 ± 0.10

Questions

8 The values of load shown in Table P2.4 are given without any indication of their uncertainties. Suggest a reason for this.

9 A student measures the radius r and the resistance R of several equal lengths of wire. The results are shown in Table P2.5. It is suggested that R and r are related by the equation:

$$R = ar^b$$

where a and b are constants.

a A graph is plotted with ln R on the y-axis and ln r on the x-axis. Express the gradient and y-intercept in terms of a and b.

b Values of r and R measured in an experiment are given in Table P2.5.

***r* / mm**	***R* / Ω**	**ln *r* / mm**	**ln *R* / Ω**
2.0 ± 0.1	175.0		
3.0 ± 0.1	77.8		
4.0 ± 0.1	43.8		
5.0 ± 0.1	28.0		
6.0 ± 0.1	19.4		

Table P2.5: Measurements for Question 9.

Copy and complete the table by calculating and recording values of ln (r / mm) and ln (R / Ω) and include the absolute uncertainties in ln (r / mm).

c Plot a graph of ln (r / mm) against ln (R / Ω). Include error bars for ln (r / mm).

d Draw the line of best fit and a worst acceptable straight line on your graph.

e Determine the gradient of the line of best fit. Include the uncertainty in your answer.

f Using your answer to part **e**, determine the value of b.

g Determine the value of a and its uncertainty.

P2.5 Conclusions and evaluation of results

In the previous experiment in P2.4, we can conclude that the extension/load for the spring in this example is (1.6 ± 0.2) cm N^{-1}. If a hypothesis is made that the extension is proportional to the load then there is enough evidence here for the conclusion to be supported, as a straight line can be drawn from the origin through all the error bars. If this is not possible then the hypothesis is not validated.

Now, suppose that the hypothesis is that the spring obeys Hooke's law and stretches by 5.0 cm when a load of 2.5 N is applied. The first part is validated for the reasons given. However, an extension of 5.0 cm for a load of 2.5 N gives a value of 2.0 cm N^{-1} for the gradient. This is clearly outside the range allowed for by the uncertainty in our measurements, and therefore the hypothesis is not supported.

REFLECTION

Make a checklist of all the important points that you are likely to forget or which led to you losing marks in any of the exercises.

SUMMARY

Planning includes:

- identifying variables that are independent, dependent and controlled
- the procedure to be followed, including a diagram, where appropriate, and the measurements to be taken
- how the measurements will be analysed, including the graph to be plotted and how the final result is calculated using the graph, for example, how the result is calculated from values of the gradient and the intercept of the graph
- extra detail, for example, how to obtain large changes in the dependent variable, an assessment of risk, a relevant safety precaution and how variables are kept constant.

Analysis of data includes:

- rearranging expressions including taking logarithms to obtain constants in expressions such as:

$$y = mx + c,\ y = ax^{n} \text{ and } y = ae^{\kappa x}$$

- plotting graphs with error bars and calculating uncertainty by the difference between the gradients of the best fit and worst acceptable lines
- calculating derived quantities with correct units and appropriate number of significant figures.

EXAM-STYLE QUESTIONS

1 Each reading on a thermometer can be made with an uncertainty of ±0.5 °C. The thermometer is used to measure a temperature rise from 20 °C to 80 °C.

What is the percentage uncertainty in the measurement of this temperature rise? **[1]**

a 0.6%

b 0.8%

c 1.7%

d 2.5%

2 The period T and the length l of a simple pendulum are measured to be $T = 1.5 \pm 0.1$ s and $l = 0.560 \pm 0.001$ m.

The formula $T = 2\pi\sqrt{\frac{l}{g}}$ is used to find the acceleration of free fall g.

What is the best estimate of the uncertainty in the value of g? **[1]**

a 0.2%

b 3%

c 7%

d 13%

3 The volume of air inside a bottle affects its resonant frequency.

a What are the dependent and independent variables? **[1]**

b Suggest one quantity to be controlled. **[1]**

c How would you produce sounds of different frequency to show that the bottle resonates? **[1]**

d How would you find the frequency of the sound that makes the bottle resonate? **[1]**

e How would you find the volume of air inside the bottle? **[1]**

f How would you change the volume of air inside the bottle while keeping all other factors constant? **[1]**

g Suggest a safety precaution involving sound. **[1]**

[Total: 7]

4 The terminal velocity of an air bubble that rises in water is affected by the size of the bubble.

a What are the dependent and independent variables? **[1]**

b Suggest a quantity to be controlled. **[1]**

c How would you measure the terminal velocity of an air bubble that rises in water? **[1]**

d How would you generate bubbles of air of different sizes in water? **[1]**

[Total: 4]

5 The count rate from a radioactive source emitting γ-radiation is inversely proportional to the square of the distance from the source. Sources emitting γ-radiation also emit α- and β-radiation and are roughly spherical with a diameter of 2 cm.

a What are the dependent and independent variables? **[1]**

b Suggest a quantity to be controlled. **[1]**

c How could you make sure that only γ-radiation is detected? **[2]**

d How would you measure the count rate? Draw a diagram of the apparatus and explain how it is used. **[2]**

e How would you make the uncertainty in the count rate as small as possible? **[1]**

f Suggest one difficulty in measuring the distance and how this difficulty may be reduced. **[2]**

g Suggest a safety precaution. **[1]**

[Total: 10]

6 The size of a small toy balloon depends on atmospheric pressure.

a What are the dependent and independent variables? **[1]**

b Suggest a quantity to be controlled. **[1]**

c Draw a diagram of an apparatus to investigate the change in size of the balloon as atmospheric pressure changes. **[2]**

d State how the pressure is changed in your apparatus and how it is measured? **[2]**

e Suggest a safety precaution. **[1]**

[Total: 7]

7 Quantities A and B have the following values: $A = 3.0 \pm 0.2$ cm, $B = 2.0 \pm 0.1$ cm. Find the value of the following expressions and their absolute uncertainties.

a AB **[1]**

b $\frac{A}{B}$ **[1]**

c A^2 **[1]**

d $A - B$ **[1]**

e $A^2 - B^2$ **[1]**

f $\sqrt{A}$ **[1]**

[Total: 6]

8 Explain how you draw the best fit line and the worst fit line on a graph and how you find the uncertainty in the intercept on the y-axis. **[2]**

Questions 9 to 13 ask you to design an experiment based on the information given. All these questions have the same marking structure, with marks allocated to the different aspects as shown.

a the procedure to be followed **[3]**

b the measurements to be taken **[5]**

c the analysis of data **[2]**

d the safety precautions to be taken **[1]**

e additional detail. **[4]**

9 The resistance R of a light-dependent resistor (LDR) varies with the distance d from a very bright source of light. It is suggested that R and d are related by the formula $R = kd^n$, where k and n are constants. Design a laboratory experiment to test this relationship. The LDR has a resistance of 50 Ω in bright light and 200 kΩ in the dark. **[15]**

10 A ruler with a small mass at one end is clamped at the other end, as shown, and oscillates up and down when plucked by hand.

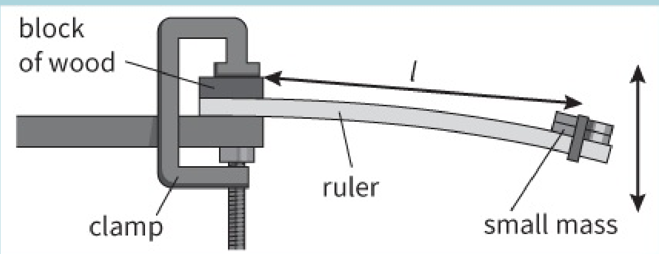

Figure P2.9

It is suggested that the period of oscillation T of the ruler is related to the length l by the relationship $T = kl^n$, where k and n are constants. Design a laboratory experiment to test this relationship and to find the value of n. **[15]**

11 A current-carrying coil produces a magnetic field. It is suggested that the magnetic field strength B at the centre of the coil is proportional to the current

I in the coil. Design a laboratory experiment that uses a Hall probe to test this relationship. **[15]**

12 A bar magnet dropped into a coil induces an e.m.f. in the coil. It is suggested that E, the maximum induced e.m.f., is proportional to v, the speed of the magnet. Design a laboratory experiment to test this relationship. You might like to look at Figure 26.24 in Chapter 26. **[15]**

13 A student has a number of different transformers of varying numbers of turns. An alternating input current to the transformer induces an output e.m.f. It is suggested that the output e.m.f. V_s is directly proportional to the frequency f of the applied current. Design a laboratory experiment to test this relationship. **[15]**

14 The period T of a simple pendulum is related to its length l by the equation:

$$T = 2\pi\sqrt{\frac{l}{g}}$$

where g is the acceleration of free fall.

a A graph is plotted with T^2 on the y-axis and l on the x-axis. Express the gradient in terms of g. **[1]**

b A student measures the time t for 10 oscillations for different lengths l. This table shows her data.

l / m	t / s	T	T^2
0.300	11.1 ± 0.1		
0.400	12.8 ± 0.1		
0.500	14.2 ± 0.1		
0.600	15.8 ± 0.1		
0.700	16.9 ± 0.1		
0.800	18.1 ± 0.1		

Table P2.6

i Calculate and record values of T and T^2, including the absolute uncertainties in T and T^2. **[3]**

ii Plot a graph of T^2 / s^2 against l / m including error bars for T^2. **[2]**

iii Draw a straight line of best fit and a worst acceptable line on your graph. **[2]**

iv Determine the gradient of your line and include the uncertainty in your answer. **[2]**

v Use your value of the gradient to determine g and include the absolute uncertainty in your value. **[2]**

vi Using your value of g and its uncertainty, calculate the value of t when the length l is 0.900 m. Include the absolute uncertainty in your answer. **[2]**

[Total: 14]

15 Readings are taken of the resistance R of a thermistor at different temperatures T. It is suggested that the relationship between R and T is $R = kT^n$, where k and n are constants.

a A graph is plotted with lg R on the y-axis and lg T on the x-axis. State the value of the gradient and the y-intercept in terms of k and n. **[2]**

b Values for T and R are shown in this table. **[4]**

T / K	R / Ω	lg (T / K)	lg (R / Ω)
273	550 ± 10		
283	480 ± 10		
293	422 ± 10		

303	370 ± 10		
313	330 ± 10		

Table P2.7

Complete the table and include absolute uncertainties in lg (R / Ω).

c **i** Plot a graph of lg (R / Ω) against lg (T / K). Include error bars. **[2]**

ii Draw the line of best fit and a worst acceptable line on your graph. **[2]**

iii Determine the gradient of your line of best fit and the uncertainty in your value. **[2]**

iv Determine the y-intercept of your graph (this is where the x-value, in this case, lg (T / K), is zero). Give the uncertainty in your value. **[2]**

v Determine values for n and k and the uncertainties in your answers. **[3]**

[Total: 17]

SELF-EVALUATION CHECKLIST

After studying the chapter, complete a table like this:

I can	See topic...	Needs more work	Almost there	Ready to move on
identify independent variables, dependent variables and variables to be kept constant	P2.2			
describe methods and procedures to vary, measure and keep variables constant	P2.2			
assess safety risks and describe precautions to reduce risk	P2.2			
understand the type of graphs to plot to produce straight lines for relationships of the form: $y = mx + c$, $y = ax^n$ and $y = ae^{\kappa x}$	P2.3			
plot lines of best fit and a worst acceptable line on a graph including the use of error bars	P2.4			
convert absolute into percentage uncertainty and vice versa	P2.4			
calculate uncertainty estimates in derived quantities and in the gradient of a graph	P2.4			
express a quantity as a value, an uncertainty estimate and a unit.	P2.4			